Lists and indexing

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Python readiness portfolio

THE CONNECTED WEEK

Teach access as a boundary on a particular list state.

Supply complete fresh initialisations, introduce -1 explicitly and preserve partial output before an error.

Keep one habit: Ask which list is active, which position exists and what later statements are skipped after failure.

MAIN LESSON · SECTION 1

Prepare the connected bridge lesson

> Check prior exposure and keep execution practice distinct from paper reasoning.

Goal and main plan

Primary target: Distinguish a **list index** from the value stored at that position

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min

> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use “Run the investigation”, “Connect your project”, “Show what you know”, “Study a complete case” and “Try a new case independently”. Give extra time for the full task set. If this week has a coding lab, use its named section; section numbers can change when extra explanations are included.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Retrieval answer guidance

Retrieval and prerequisite guidance

Without opening the old lesson, explain one key idea from Loops and trace tables. Give an example and a mistake that the idea helps you avoid.

Look for: Trace variable values before and after each loop iteration

Reference example, not the only acceptable wording: 2, 3 and 4; the stop value is excluded. The body runs three times.

Boolean conditions

Without opening the old lesson, explain one key idea from Boolean conditions. Give an example and a mistake that the idea helps you avoid.

Look for: Translate stated requirements into comparisons

Reference example, not the only acceptable wording: (minutes <= 12) and quiet. At most includes equality, so <= is required; the nonnegative integer and Boolean **input** contract still applies.

From a paper algorithm to Python

Without opening the old lesson, explain one key idea from From a paper algorithm to Python. Give an example and a mistake that the idea helps you avoid.

Look for: Translate a short paper calculation into a Python script

Reference example, not the only acceptable wording: packs = 4; per_pack = 6; total = packs * per_pack; print(total), each on its own line. The first two bind quantities, the third computes 24 and the fourth displays 24.

Assignment alone produces no displayed **output** in this script.

Teach **list**s as ordered collections, with **index** and value kept distinct. The first exercises use nonnegative indices, although Python also permits negative **index**ing.

Reference: https://docs.python.org/3/tutorial/introduction.html#**list**s. Do not imply that updating a record changes or verifies the physical object represented.

Materials and readiness

Python 3 editor and **list**s.py

Four position cards numbered 0 through 3 and movable value cards

Blank trace table and separate predicted and observed **output** columns

0, 1 and 2; three iterations. The stop value 3 is excluded.

End goal

Trace **list**s and **index**ing and explain the changed program before running it.

▼ Earlier wording, alternatives and picture transcripts

Teach the learning cycle

Loops and trace tables

Outcomes

Distinguish a **list index** from the value stored at that position

Read and update an **item** using a valid **index**

Trace a loop through **list** values and test an empty **list**

Preparation

Prerequisite answer

End goal for the pictorial case

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

MAIN LESSON · SECTION 2

Teach list state and access boundaries together

> The negative-index extension is explicit and only valid for an existing item.

Read positions, types and main update

A fictional supply shelf has three trays with counts 4, 7 and 4 in that order. Store them as counts = [4, 7, 4]. Square brackets enclose the **list** and commas separate its **item**s.

The order matters: the first and third trays contain equal values but occupy different positions.

Python counts positions from zero: counts[0] is first, counts[1] second and counts[2] third. len(counts) is 3 for three **item**s; it is not the last nonnegative **index**. Start with nonnegative **index**ing, then use this lesson's pictured extension: **index** -1 reads the final **item** of a

nonempty list. Both positive and negative **item** access require an existing position; an empty list has no last **item**.

The printed lines are 3, 4, 7 and [4, 9, 4]. The assignment counts[1] = 9 replaces the **item** at **index** 1; it does not add a fourth **item** or change the two other values. The **length** stays 3.

A recorded count can be changed by code without proving that the physical tray really contains that many objects.

Trace a fresh-list sum

```
counts = [4, 7, 4]
total = 0
for count in counts:
    total = total + count
print(total)
```

Item read as count	total before	total after
4	0	4
7	4	11
4	11	15

This fresh script prints 15. for count in counts supplies the stored values, not the indices. Both copies of 4 are **process**ed because both are **item**s. For an empty **list** written [], the body runs zero times and total stays 0.

For [7], it runs once and prints 7.

Read the pictured list and scenario labels

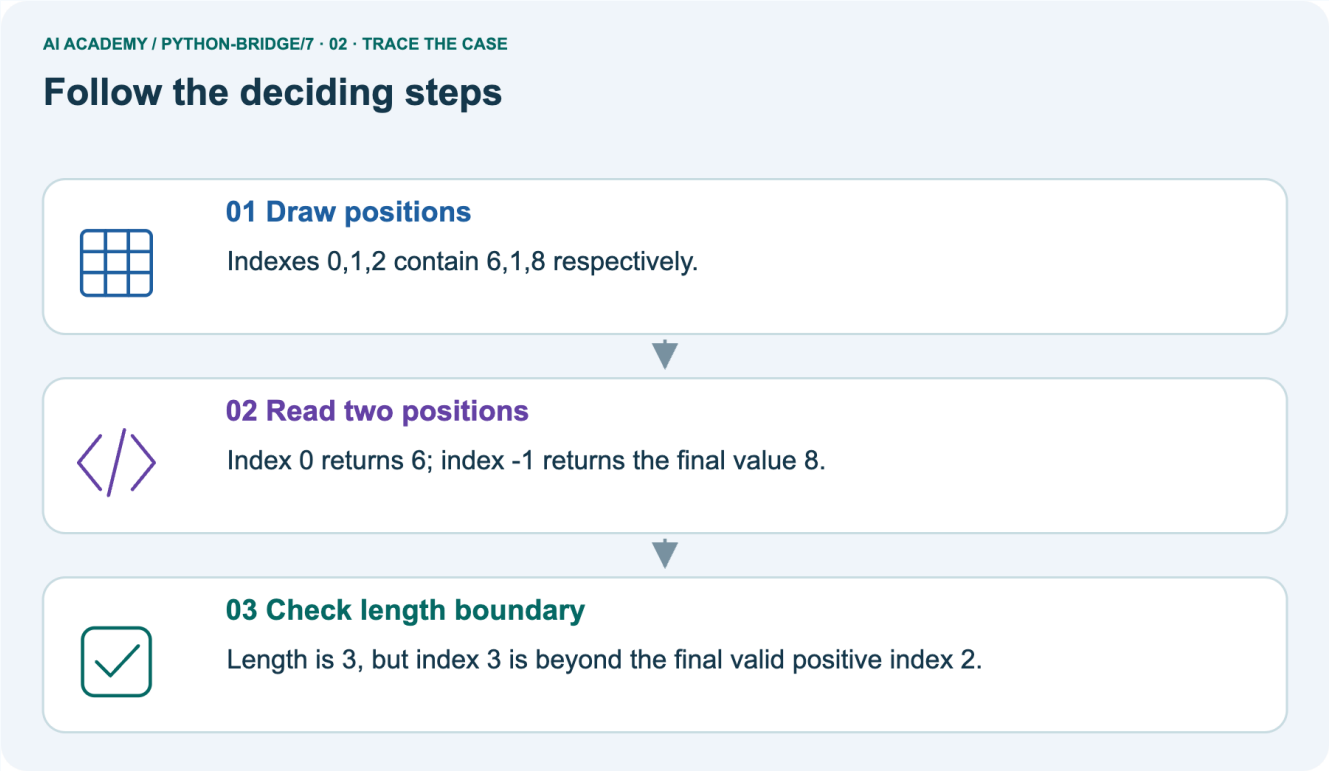
Indexes are positions, not values. A **list** of shelf counts is read by position.

Python 3: counts=[6,1,8]; print(counts[0]); print(counts[-1]); print(len(counts)).

Label these independent initialisations: main list-read script starts [4,7,4] and updates index 1 to 9, producing [4,9,4]. The fresh sum script restarts [4,7,4], so its total is 15. W2 again restarts [4,7,4] but changes index 0 to 6, yielding [6,7,4]. The picture separately uses [6,1,8]. Never silently carry one script’s mutation into another scenario.

For any nonempty list, -1 reads its final item; an empty list has no valid item access.

Trace positions and inspect the original graphic



Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / PYTHON-BRIDGE/7

Zero-based list positions



Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Draw positions	Indexes 0,1,2 contain 6,1,8 respectively.
2. Read two positions	Index 0 returns 6; index -1 returns the final value 8.
3. Check length boundary	Length is 3, but index 3 is beyond the final valid positive index 2.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

List positions begin at zero. A negative **index** counts from the end; the **list length** counts entries and is not itself a valid positive **index** for the final entry.

```
counts = [6, 1, 8]
print(counts[0])
print(counts[-1])
print(len(counts))
```

Predict the changed invalid access

Change: Replace the second print with print(counts[3]).

Prediction: The first print shows 6, then **IndexError** stops the script before the **length** print.

Reason: The attempted position does not exist.

State source-accuracy and empty limits

A valid position does not establish that the stored count is accurate; empty **list**s also have no final element to read.

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

▼ Additional guided key

For [2, 0, 5], totals become 2, 2, 7. **Output** is 7. Zero is still an **item** and its iteration still executes. Exit: len([8, 3]) is 2, but **index** 2 is outside the **list**. Last nonnegative **index** is 1, whose value is 3.

An empty **list** has no last **item**, so a **length** check is required before access.

▼ Earlier wording, alternatives and picture transcripts

Term	Meaning
List	An ordered collection of items
Index	An integer position used to access an item

Term	Meaning
Length	The number of items in a collection
Item	One value stored in a collection
IndexError	An error raised when an index is outside a sequence

Positions and stored counts

TRAY POSITION	INDEX	VALUE
First	0	4
Second	1	7
Third	2	4

Equal stored values do not imply equal positions. There is no item at index 3.

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

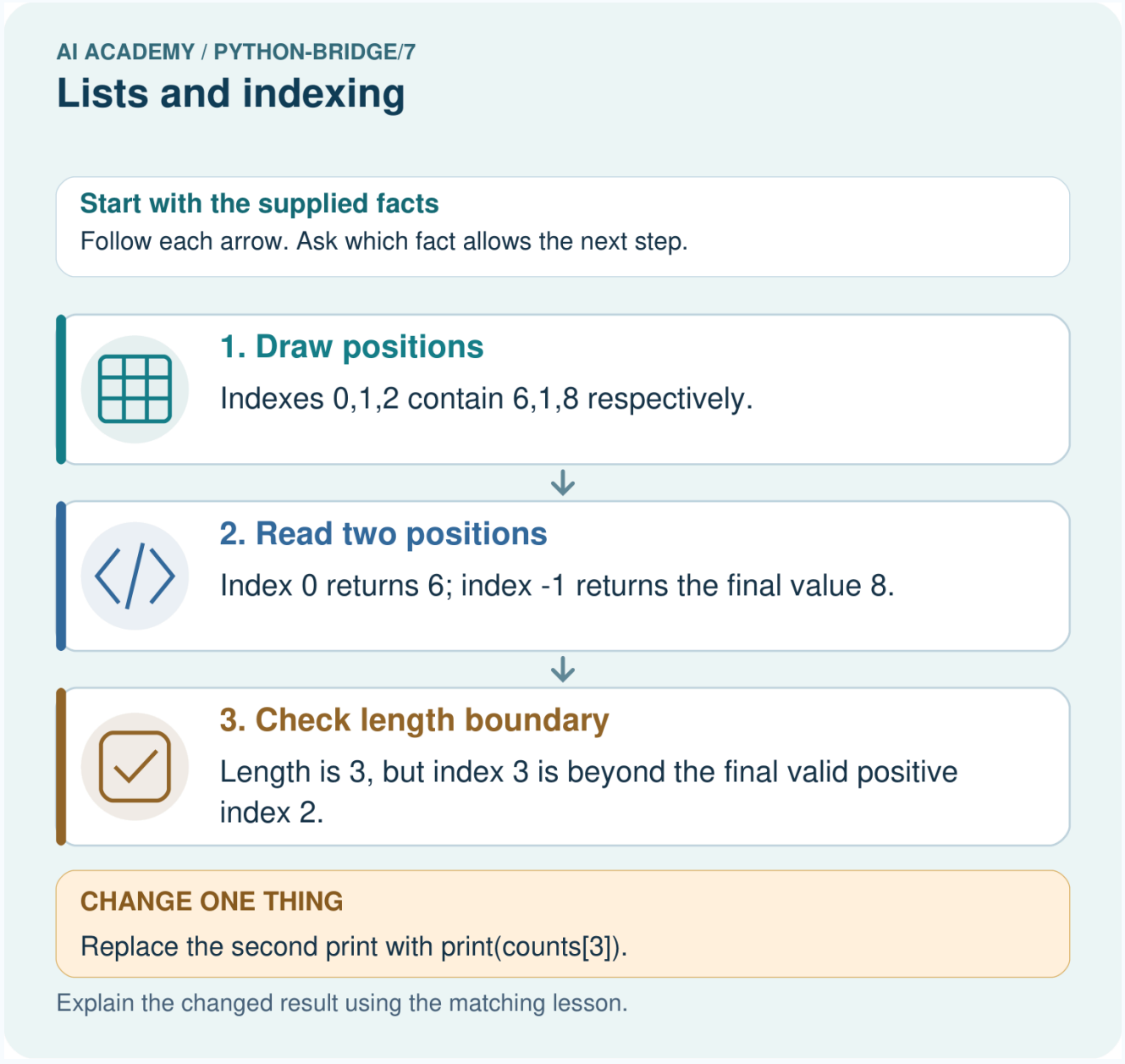
> **Change only the list to [2, 0, 5]. Trace the three additions and state the final output. Explain why the middle item is processed even though its value is zero.**

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Lists and indexing

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Change one thing in the pictured case

Ask while showing the picture

MAIN LESSON · SECTION 3

Guide the trace, then fade the help

> Collect predictions before execution or feedback.

Main completion task and key

Completion task: Start with counts = [4, 7, 4]. Run counts[0] = 6. Write the resulting **list**, its **length** and the value at **index** 2. Predict before running.

Teacher check: The result is [6, 7, 4]; **length** remains 3; **index** 2 still holds 4. Only the first **item** was replaced.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual misconception check

Use the worked case for **List**s and **index**ing. Proposed explanation: "**Length** is the last nonnegative **index**". Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: **Length** counts **item**s; the last nonnegative **index** of a nonempty **list** is one less.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictorial question and changed-case key

Label the three **index** positions. Predict all original **output**s, then name the first **output** and stopping point after using **index** 3.

Original **output**s are 6,8,3. After the change, 6 prints first, counts[3] raises **IndexError**, and the third print does not execute.

Changed-case reasoning: The first print shows 6, then **IndexError** stops the script before the **length** print. The attempted position does not exist.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

▼ Support and alternative plan

Time or stage	Teacher action
Session 1 start to 7 minutes	Retrieve range(3) and its excluded stop.
7 to 20 minutes	Place value cards under position cards and complete W1.
20 to 33 minutes	Run the worked read and update script, then complete W2.
33 to 45 minutes	Trace the value loop and answer the guided question.
Session 2 start to 8 minutes	Retrieve the distinction between length and final index.
8 to 20 minutes	Predict the faulty access before running; use W3 to repair value versus index confusion.
20 to 35 minutes	Collect W4 independently, keeping the replacement step separate from empty-list summation.
35 to 45 minutes	Discuss W5 and use the exit to diagnose indexing assumptions.

Use the worked example in Sections 3 and 6. Collect a **prediction** before revealing its result; ask the learner to point to each intermediate fact.

Keep position cards fixed while moving value cards. Permit an oral trace and provide a blank grid.

Extension: write a guarded final-**item** reader that prints Empty when len(readings) == 0, otherwise prints readings[len(readings) - 1]. Test [] and [6].

▼ **Earlier wording, alternatives and picture transcripts**

Compact alternative: this two-session sequence covers selected core tasks. Use the three-session weekly plan for the full lesson, independent transfer, feedback and project work; allow extra time for extensions.

Reduce help deliberately

Individual reasoning check

Pictorial practice question

Reasoned teacher answer

MAIN LESSON · SECTION 4

Run the complete supplied investigation

> Preserve expected, observed and paper-trace records.

Supplied case materials and checked guidance

W2 list [6,7,4], length 3, index 2 value 4. Fresh sum 15; empty 0; singleton 7. In a values loop, counts[count] first attempts index 4 and fails for [4,7,4]; print(count) uses the value directly. Pictured outputs 6,8,3; changed second access prints 6 then raises IndexError, skipping length. Selected outputs 9,2,3; changed second access prints 9 then IndexError, skipping final length.

Guarded [] reports no last item, [0] prints 0. Earlier W4 retains its own values.

Diagnose misconceptions

Length is the last nonnegative **index**: **Length** counts **item**s; the last nonnegative **index** of a nonempty **list** is one less.

A loop variable always contains an **index**: A direct **list** loop supplies stored values.

Updating one **item** grows the **list**: Assignment at an existing **index** replaces that **item**; **length** is unchanged.

Repeated values are **process**ed only once: Each position contributes one **item**, even when values repeat.

Use Section 9 for the **evidence** ledger. Add a boundary or missing-**input** case and record the expected response before checking. More successful repeats of one case do not establish wider coverage.

Choose accessible support

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.

What the response shows	Next teaching action
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

▼ **Earlier wording, alternatives and picture transcripts**

Match the support to the first error

MAIN LESSON · SECTION 5

Assess the selected bridge transfer task

> If a learner has seen this exact case, use a fresh parallel input and record exposure/support.

Selected independent task and checked key

Selected case: Predict the three **output**s. If the second print uses values[3] instead, where does execution stop and which later **output** is not produced?

```
values = [4, 9, 2]
print(values[1])
print(values[-1])
print(len(values))
```

Outputs are 9,2,3. With values[3], the first print produces 9, then **IndexError** occurs; the final **length** print is not reached. Positive valid indices are 0,1,2.

Judge trace and support

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Give one repair action

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

W1 Separate index and value

For counts = [4, 7, 4], give the **length**, valid nonnegative indices and value at **index** 1. Explain how the two **item**s with value 4 differ.

Length 3; indices 0, 1, 2; counts[1] is 7. The equal values occupy different positions, indices 0 and 2.

W2 Trace an update

Start with counts = [4, 7, 4]. Run counts[0] = 6. Write the resulting **list**, its **length** and the value at **index** 2. Predict before running.

The result is [6, 7, 4]; **length** remains 3; **index** 2 still holds 4. Only the first **item** was replaced.

W3 Repair a position mistake

A learner loops with for count in counts and tries to read counts[count] for counts = [4, 7, 4]. Identify the first attempted **index** and error. Explain how to print each value directly.

The first count is 4, so counts[4] raises **IndexError**. The loop variable already holds each value. Use an indented print(count) to print 4, 7 and 4 on separate lines.

W5 Keep the information you need

Compare [1, 5, 2] and [5, 1, 2]. State their totals and first **item**s. Would storing only the total answer which first **item** was larger? Explain.

Both total 8. Their first **item**s are 1 and 5, so the second **list** has the larger first **item**. A total alone discards the order and cannot answer this question.

▼ Earlier wording, alternatives and picture transcripts

Guided and exit questions

For [2, 0, 5], totals become 2, 2, 7. **Output** is 7. Zero is still an **item** and its iteration still executes. Exit: len([8, 3]) is 2, but **index** 2 is outside the **list**. Last nonnegative **index** is 1, whose value is 3.

An empty **list** has no last **item**, so a **length** check is required before access.

W1 Separate index and value

W2 Trace an update

W3 Repair a position mistake

W4 Process new data independently

Start with readings = [3, 0, 3, 2]. Replace the second **item** with 5 using its **index**. Write a loop that adds all updated readings and prints one final total. Trace every update of the total.

Then test the summing part alone on [] and [6].

Use readings[1] = 5, giving [3, 5, 3, 2]. Initialize total = 0 before for reading in readings, add reading inside, and print outside. Running totals are 3, 8, 11, 13; final 13. Summing [] gives 0 and [6] gives 6.

The replacement line itself requires at least two **item**s.

W5 Keep the information you need

Independent assessment and fresh retry

W4 must use **index** 1 for the second **item**, retain all four **item**s, produce the full trace and **output** 13, and obtain 0 and 6 when testing only summation.

After support, retry [2, 2, 1], replace the first **item** with 4, and sum: updated **list** [4, 2, 1], totals 4, 6, 7. Then sum [] and [5] for **output**s 0 and 5. Record support before judging independence.

Assess independent transfer

Selected case answer

Feedback that the learner uses

MAIN LESSON · SECTION 6

Keep the portfolio, support and delayed check

> Track code execution separately from a correct paper trace.

Project evidence

Add a labelled **list**s and **index**ing **evidence** sheet to Python readiness portfolio. Show how you can distinguish a **list index** from the value stored at that position.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

For Python readiness before the Responsible Plant Classifier, retain a trace of the values, the predicted **output**s and the changed-case explanation. Use actual Python 3 results to check the **prediction** after reasoning first.

Support and selected assessment guidance

Keep position cards fixed while moving value cards. Permit an oral trace and provide a blank grid.

Extension: write a guarded final-**item** reader that prints Empty when len(readings) == 0, otherwise prints readings[len(readings) - 1]. Test [] and [6].

Selected independent answer: Outputs are 9,2,3. With values[3], the first print produces 9, then IndexError occurs; the final length print is not reached. Positive valid indices are 0,1,2. Keep earlier W4 transfer as additional practice; do not substitute its numbers or key for this selected task. Record prior exposure and use a fresh parallel case after support.

Delayed return

Suggested return points in this level: python-bridge/8, python-bridge/10. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	

Record	Teacher evidence
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Bridge to the next topic

Week 8 groups reusable instructions in functions and supplies values through arguments.

Retain the project ledger and assess a changed case after support. Check the relevant prerequisite before carrying this component forward.

▼ Earlier wording, alternatives and picture transcripts

Plan the delayed return

Project evidence from this case

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Alternative planning checklist**

REFERENCE · SECTION 7

Reference · Alternative planning checklist

> Use alongside the main route without repeating the teaching cycle.

Planning and assessment reference

Teaching target: Distinguish a **list index** from the value stored at that position

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use “Run the investigation”, “Connect your project”, “Show what you know”, “Study a complete case” and “Try a new case independently”. Give extra time for the full task set. If this week has a coding lab, use its named section; section numbers can change when extra explanations are included.

Before class, solve the supported practice: Start with counts = [4, 7, 4]. Run counts[0] = 6. Write the resulting **list**, its **length** and the value at **index** 2. Predict before running.

Reasoned practice answer: The result is [6, 7, 4]; **length** remains 3; **index** 2 still holds 4. Only the first **item** was replaced.

Check every learner before discussion: Use the worked case for **List**s and **index**ing. Proposed explanation: “**Length** is the last nonnegative **index**”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: **Length** counts **item**s; the last nonnegative **index** of a nonempty **list** is one less.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: Replace the second print with print(counts[3]). Require a **prediction** and a reason before comparing. Expected result: The first print shows 6, then **IndexError** stops the script before the **length** print.

Reason: The attempted position does not exist.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint: Python readiness portfolio. Keep runnable code, **prediction**s, actual **output**s, one repaired error and an explanation of a function on changed **input**s.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

▼ **Reference · Feedback and evidence records**

REFERENCE · SECTION 8

Reference · Feedback and evidence records

> Keep individual first attempts separate from supported rehearsal.

Feedback and repair

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and **index**ing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

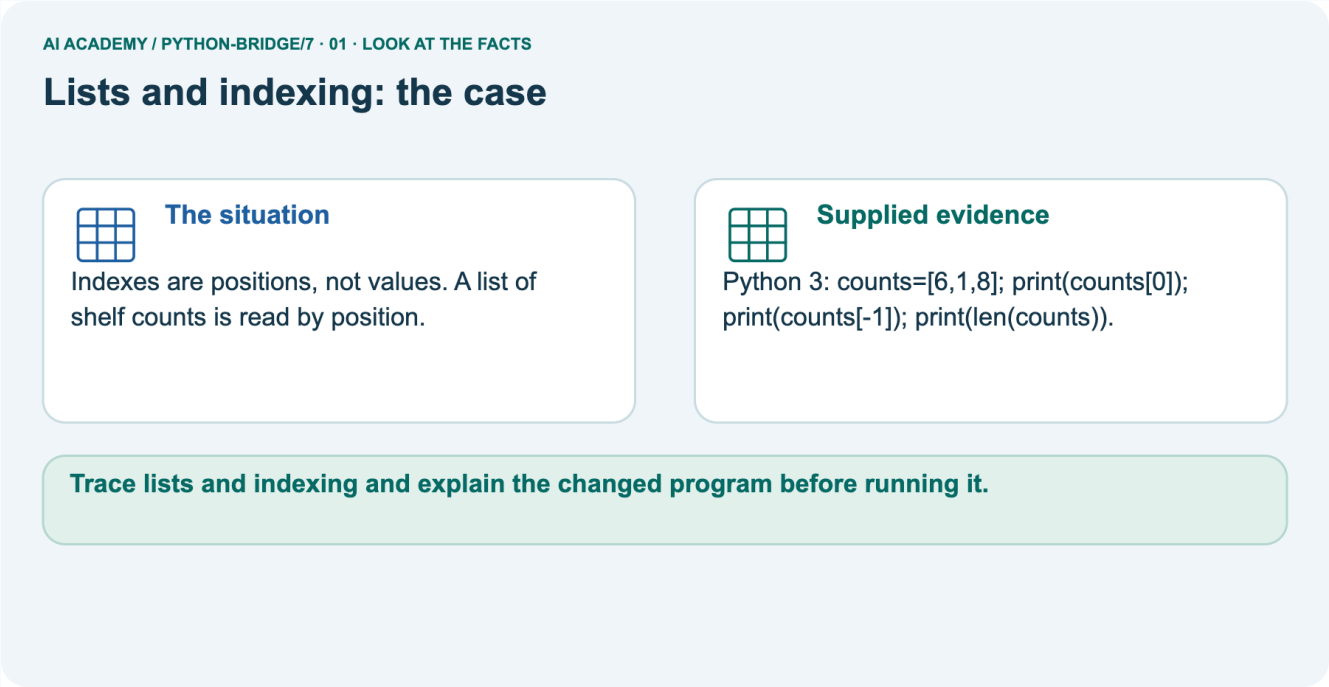
▼ **Reference · Teach with the six pictures**

REFERENCE · SECTION 9

Reference · Teach with the six pictures

> Use the relevant picture for a specific reasoning move.

01. Lists and indexing: the case



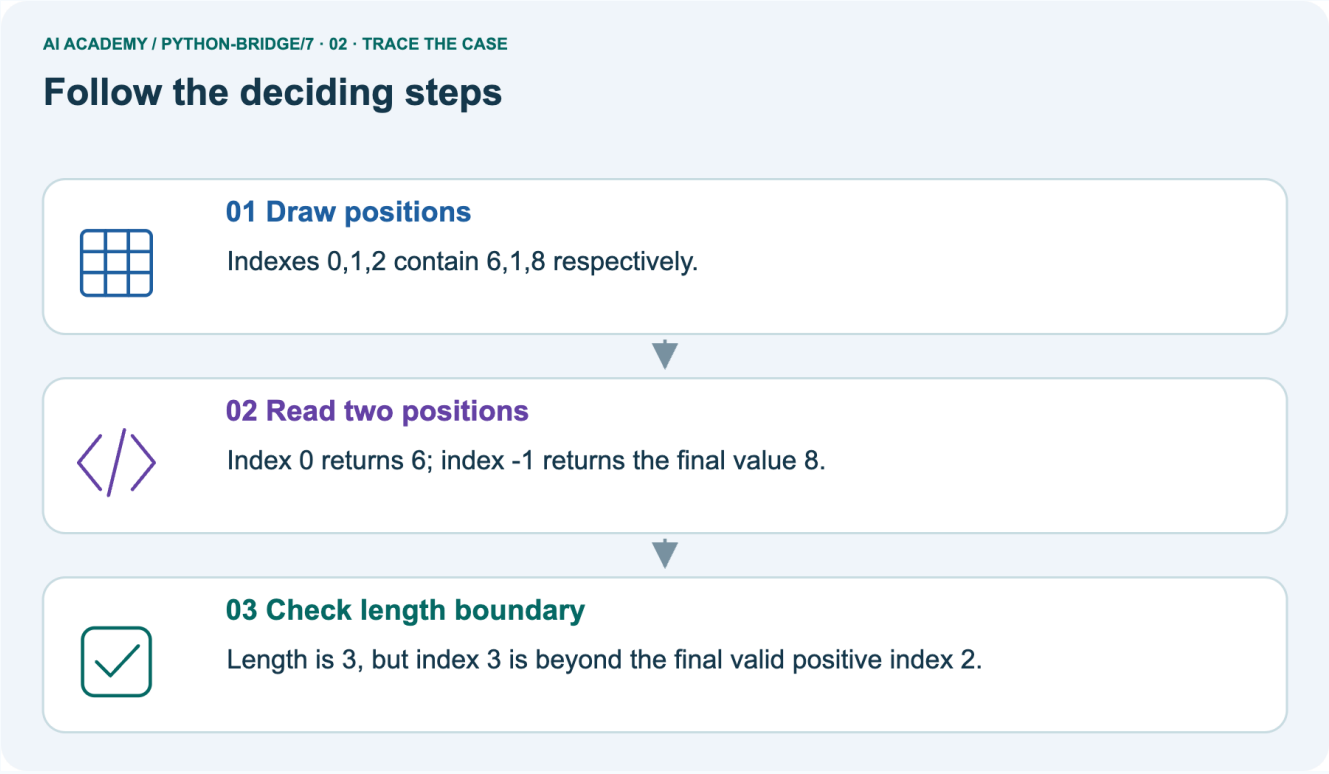
Follow the labelled picture with the matching explanation. Open the vector diagram

Indexes are positions, not values. A **list** of shelf counts is read by position. Supplied facts: Python 3: counts=[6,1,8]; print(counts[0]); print(counts[-1]); print(len(counts)).

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Draw positions: **Index**es 0,1,2 contain 6,1,8 respectively. → Read two positions: **Index** 0 returns 6; **index** -1 returns the final value 8. → Check **length** boundary: **Length** is 3, but **index** 3 is beyond the final valid positive **index** 2.

Point to each numbered stage and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Zero-based list positions



Follow the labelled picture with the matching explanation. Open the vector diagram

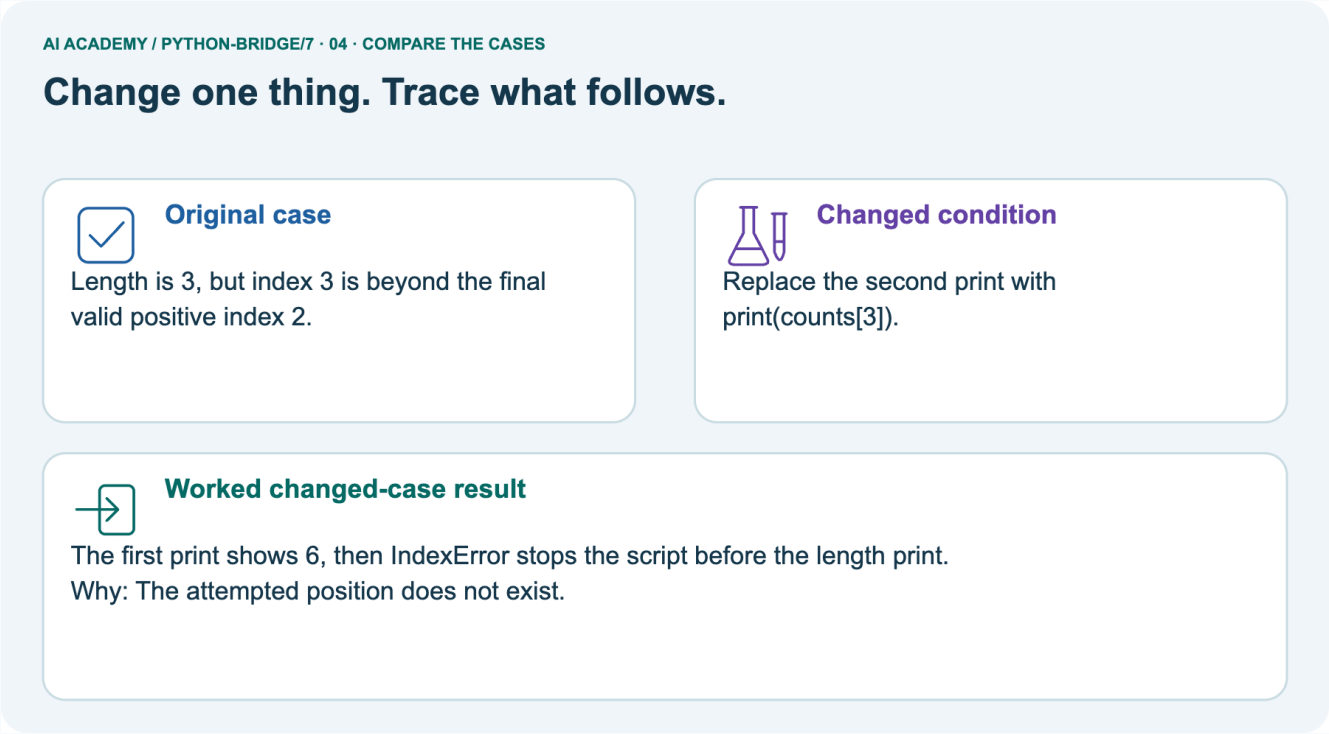
Zero-based **list** positions. Stored counts: **Index** 0 = 6, **Index** 1 = 1, **Index** 2 = 8.

Read the labels and values. Explain the relationship the model shows.

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Check row/column meanings, units and denominators before discussing a difference. Supplied toy values are not real class or model results.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: **Length** is 3, but **index** 3 is beyond the final valid positive **index** 2. Changed condition: Replace the second print with print(counts[3]).

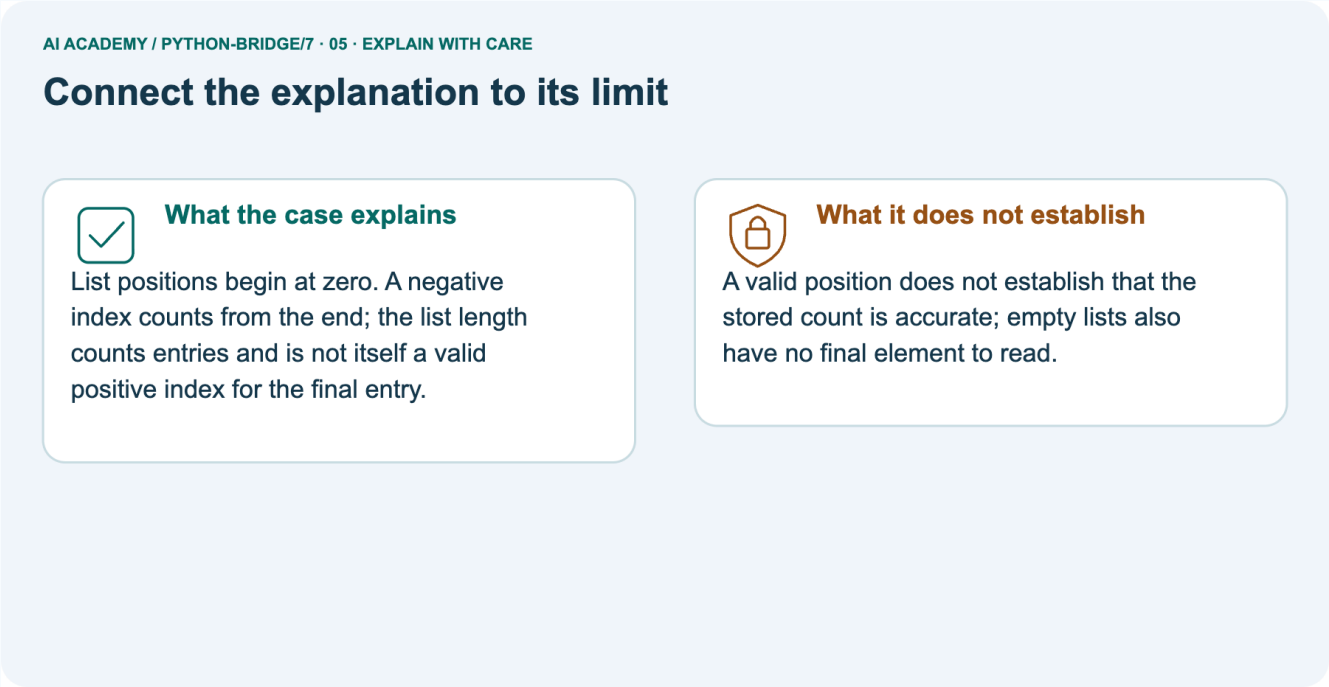
Expected result: The first print shows 6, then **IndexError** stops the script before the **length** print. Reason: The attempted position does not exist.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: The attempted position does not exist.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

List positions begin at zero. A negative **index** counts from the end; the **list length** counts entries and is not itself a valid positive **index** for the final entry.

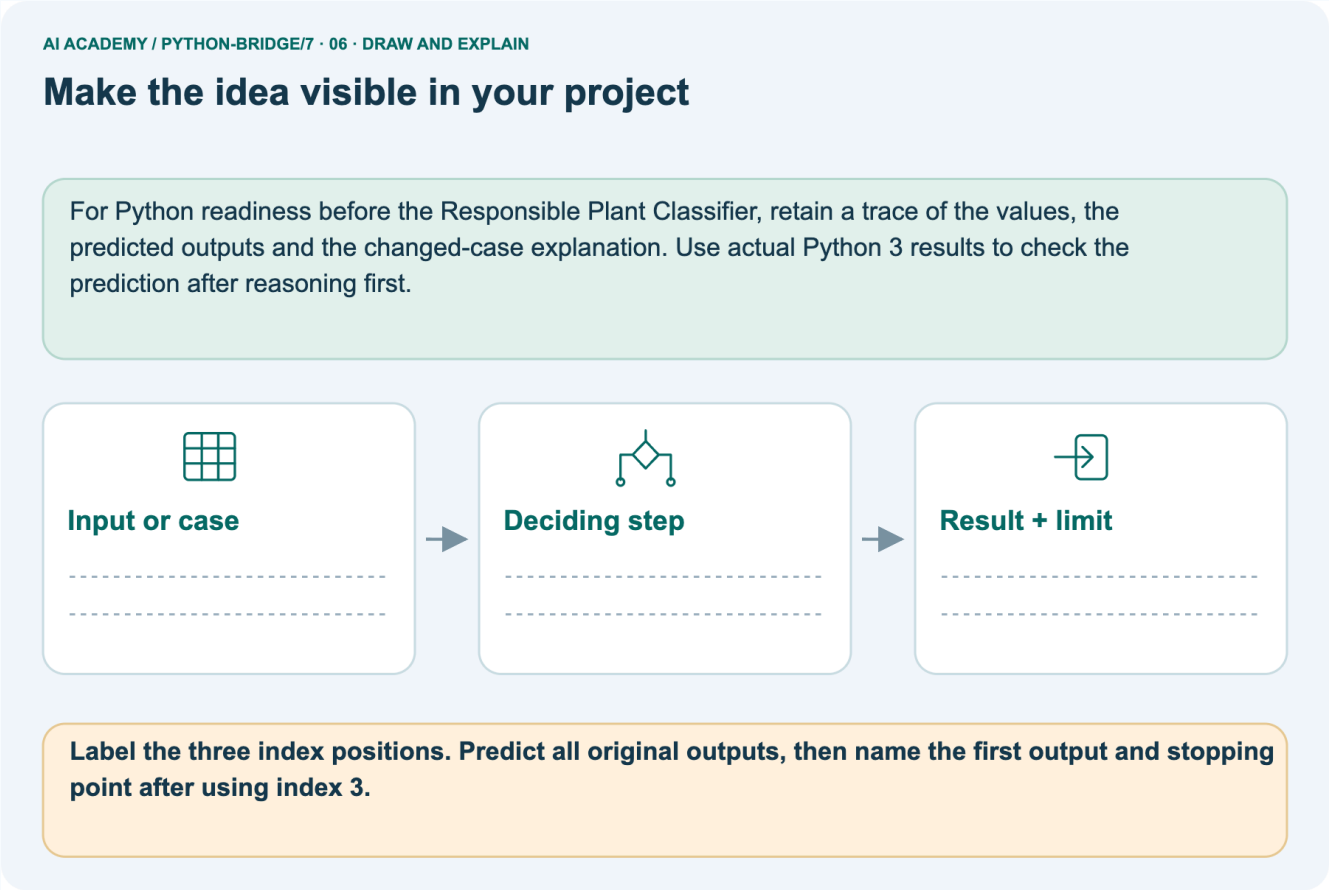
Limit: A valid position does not establish that the stored count is accurate; empty **list**s also have no final element to read.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: A valid position does not establish that the stored count is accurate; empty **list**s also have no final element to read.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For Python readiness before the Responsible Plant Classifier, retain a trace of the values, the predicted **output**s and the changed-case explanation. Use actual Python 3 results to check the **prediction** after reasoning first.

Draw your **input**, deciding step, result and limit. Supported question: Label the three **index** positions. Predict all original **output**s, then name the first **output** and stopping point after using **index** 3.

Draw your project trace and justify the deciding step using **evidence**.

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: Original **output**s are 6,8,3. After the change, 6 prints first, counts[3] raises **IndexError**, and the third print does not execute.

▼ Earlier wording, alternatives and picture transcripts

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Lists and indexing: the case

02. Follow the deciding steps

03. Zero-based list positions

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.